

Zihao Zhou

WORLD MODELS · EMBODIED AI · ROBOT LEARNING

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Research Interests

World models for embodied agents, centred on **memory and long-horizon context**: how an agent retains and retrieves task-relevant history, and how it improves from outcomes it predicts and observes rather than from demonstrations alone. I have attacked this from three sides — memory for a world model driving dexterous manipulation, long-term memory for language agents, and injecting structured world knowledge into generative models so that it transfers from the visual to the action domain.

Education

Beihang University (BUAA)

M.ENG. IN COMPUTER SCIENCE AND TECHNOLOGY

• Excellent Engineer Program. Advisor: Prof. Di Huang.

Beijing, China

Sep. 2026 – Jun. 2029 (expected)

Beihang University (BUAA)

B.ENG. IN COMPUTER SCIENCE AND TECHNOLOGY

• Weighted average **89/100**. **Third Prize**, Feng Ru Cup (BUAA's university-wide research competition); **municipal-level Outstanding Completion**, National Undergraduate Innovation and Entrepreneurship Training Program.

Beijing, China

Sep. 2022 – Jun. 2026

Publications

† denotes joint first authorship; * denotes corresponding author.

[1] Motus2: A Self-Evolving General World Model for Dexterous Manipulation

ICLR 2027, under review

HONGZHE BI†, ZIHAO ZHOU†, YIHANG TANG†, JINGRUI PANG†, SHUHE HUANG†, ET AL., JUN ZHU*

arXiv:2608.30237 | Project page

• Successor to *Motus: A Unified Latent Action World Model* (CVPR 2026). One set of shared weights exposes a policy, an action-conditioned simulator and a value model in a closed decision-and-learning loop, so the policy improves from its own failed and suboptimal interactions. On a biomimetic platform with stereo vision, dual arms, dual dexterous hands and tactile sensing, average success over five dexterous tasks rises **0%** → **51%** → **84%** across initializations that differ only in pretraining stage under identical SFT and evaluation; $\pi 0.5$ and the video-model initialization both score **0%**, placing the floor at the difficulty of 20+ DoF dexterity rather than at baseline choice.

[2] EruDiff: Refactoring Knowledge in Diffusion Models for Advanced Text-to-Image

ECCV 2026

Synthesis

XIEFAN GUO, XINZHU MA, HAOXIANG MA, ZIHAO ZHOU, DI HUANG*

arXiv:2603.20828 | Code

• Aligns the knowledge distribution of implicit prompts with explicit anchors (DK-DM) and applies negative-only reinforcement learning (NO-RL) to correct rendering bias. Sets a new state of the art for diffusion-only models on the WISE world-knowledge benchmark.

[3] Exploiting Intra-Chip Locality for Multi-Chip GPUs via Two-Level Shared L1 Cache

Journal of Systems Architecture

XIANGRONG XU, LIANG WANG*, LIMIN XIAO*, LEI LIU, ZIHAO ZHOU, YUANQIU LV, LI RUAN, XILONG XIE, MENG HAN, XIAOJIAN LIAO

vol. 167, 2025 | DOI

• TLS-Cache improves IPC by 30.2% on average over a 4-chip GPU baseline with private L1 caches — and by 8.8% over the remote-data-cache prior art — without adding cache capacity.

Research Experience

GensPI (ShengShu; creators of Vidu and Motus, CVPR 2026, 1.2k GitHub stars)

Beijing, China

RESEARCH INTERN · MEMORY, REAL-ROBOT MBRL & TRAINING INFRASTRUCTURE

Jul. 2026 – Present

- **Motus2** [1] (ICLR 2027, under review) — a self-evolving world model whose shared weights act as policy, simulator and evaluator in one closed loop. **Joint first author** (2nd of 5): **owned the memory component** and **MBRL on the real robot**, and accelerated training across the project's MBRL, tactile and main experiments.
- **Working memory (owner)**. Made accessible history an explicit design axis in a partially observable setting, and built from scratch both long-history replacements for the model's bounded sliding window — full retention by global autoregression against compressed persistence by MemoryWAM-style memory tokens — with the probe environments that isolate recall from general competence. Compression lost decisively, **57.5% vs 25.0%** on the real robot and **78% vs 52%** in simulation, identifying unbounded KV growth as the current price of reliable recall.
- **Real-robot MBRL**. Carried value-guided MBRL from simulation onto the physical platform and owned the result there: success rose **65.0% to 72.5%** on *Put Phone* and *Multi-Finger*, and to **75.0%** with Best-of- N planning.
- **Training infrastructure**. Rebuilt MBRL training on Ray in place of the original synchronous loop: **29 s** → **6 s** per step on one node, **2 s** on four. Separately accelerated the tactile and main experiments, each a distinct codebase, from **18 s to 2.4 s** and **2.4 s to 1.8 s** per step. Also stood up the simulation branch used for the memory probes.
- **Robot inference stack** (sole author). Wrote the inference stack for the 20-DoF Wuji Hand 2 and 22-DoF Sharpa Wave dexterous hands, cutting per-chunk execution drift from **21 mm to 0.9 mm**, and maintained the data-collection stack behind the project's real-robot datasets.

- **ScaleEdit (undergraduate thesis, sole investigator)** — physically scale-consistent object replacement in image editing, with the problem formulation, benchmark and method all original and uncoauthored.
- Editing models place a replacement object at a plausible-looking but physically wrong size, and no existing benchmark measures it.
- Quantified the distortion, then assembled an object-replacement benchmark grounded in real-world size priors, with a consistency protocol combining 3D geometry and VLM judging.
- Proposed scale-aware supervised fine-tuning, lifting FLUX.2-Klein-9B from **45.8 to 60.4** overall — **open-source state of the art**, second only to GPT-Image-2 (61.1).
- **EruDiff** (ECCV 2026) [2] — built **Knowledge-10K**, the 10K *text-only* corpus behind the paper’s WISE results, pairing each implicit world-knowledge prompt with an explicit anchor and enforcing uniqueness and visual verifiability through an LLM audit chain. Fine-tuning FLUX.1-dev on it raised WISE from **0.50 to 0.64**, the state of the art among diffusion-only models.

- **Mahjong VLA — project owner, two-person team.** Built the full single-view closed loop (tile perception → state tracking → policy → VLA control) that drove a bimanual **ALOHA** robot through a live two-player match — a long-horizon, imperfect-information task on real hardware.
- *Perception.* Trained a YOLO tile detector on 5K hand-annotated frames, then fine-tuned a Qwen3-VL backbone on 20K position-ordered pseudo-labels — **91.2%** exact match on full-hand tile sequences.
- *Policy.* Cold-started with SFT on DeepSeek chain-of-thought over high-discriminability decision points from Mortal self-play, then applied **GRPO** in a Mortal-derived RL environment, raising the first-place rate from **18.3% to 31.7%** over 448 half-games.
- *Control.* Routed two LoRA adapters over a π **0.5** VLA policy to separate “grasp” from “discard”, mitigating multi-task forgetting and reaching **89.3%** action success on the real robot.
- **MileTalk — a long-term-memory benchmark for in-vehicle agents.** A user simulator drives multi-day dialogues and the agent is scored, inside a real tool-calling pipeline, on whether it *remembers*, *answers* and *acts* correctly; 7 memory capabilities, benchmarked across full-context, RAG, Mem0 and Letta.
- **Adopted as the company’s internal memory evaluation standard** for the SuperMate in-vehicle assistant.
- Also owned the vision-language navigation module for **SafeClaw** (MuJoCo environment and A* baseline) and shipped a bilingual **wake-word detection** pipeline into the company’s robot product line.

- Developed the instruction-execution, memory and scheduling modules of **Swift-GPUSim** and **accel-mcm**, simulators for multi-chip GPUs.
- Independently designed the two-level crossbar published in [3], which lets SMs share L1 data within a chip and so cuts cross-chip traffic, at **0.57% area** and **0.92% power** overhead under 28 nm synthesis.

Skills

Research Areas	World models, model-based RL, vision-language-action policies, agent memory, diffusion models
Robotics	ALOHA (bimanual), MuJoCo, π 0.5
Models & Methods	FLUX, Qwen-Image, Qwen3-VL, YOLO, LoRA, GRPO, SFT, CoT distillation
Systems & Programming	Python, C/C++, CUDA; asynchronous distributed training, multi-GPU scheduling (Ray + Redis), inference optimisation (ONNX / TorchScript)
Languages	Chinese (native), English